

Website Traffic Analysis Report

Comprehensive Data Science Implementation & Predictive Modeling

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Website Traffic Analysis

1. Data Collection & Preprocessing
2. Create Visualizations
3. Model Selection
4. Model Pipeline And Training
5. Model Evaluation And Saving
6. Conclusions

Step 1 - Importing Libraries

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score
```

```
In [2]: # Load and preview statistics
df = pd.read_csv('website_wata.csv')
df.describe()
```

Out[2]:

	Page Views	Session Duration	Bounce Rate	Time on Page	Previous Visits	Conversion Rate
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000
mean	4.907000	3.053020	0.424223	4.058922	4.509000	0.952857
std	2.169725	3.122988	0.241108	2.810246	2.846572	0.136784
min	0.000000	0.000476	0.010257	0.027092	0.000000	0.341501
25%	3.000000	0.817369	0.213194	1.957940	2.000000	1.000000
50%	5.000000	2.024048	0.422743	3.409312	4.000000	1.000000
75%	6.000000	4.273986	0.641032	5.499930	7.000000	1.000000
max	14.000000	20.425786	0.839938	17.753365	9.000000	1.000000

Step 2 - Data Preprocessing

In
[3]:

```
# Checking for missing values
df.isnull().sum()
```

Out[3]:

```
Page Views      0
Session Duration 0
Bounce Rate     0
Traffic Source  0
Time on Page    0
Previous Visits  0
Conversion Rate  0
dtype: int64
```

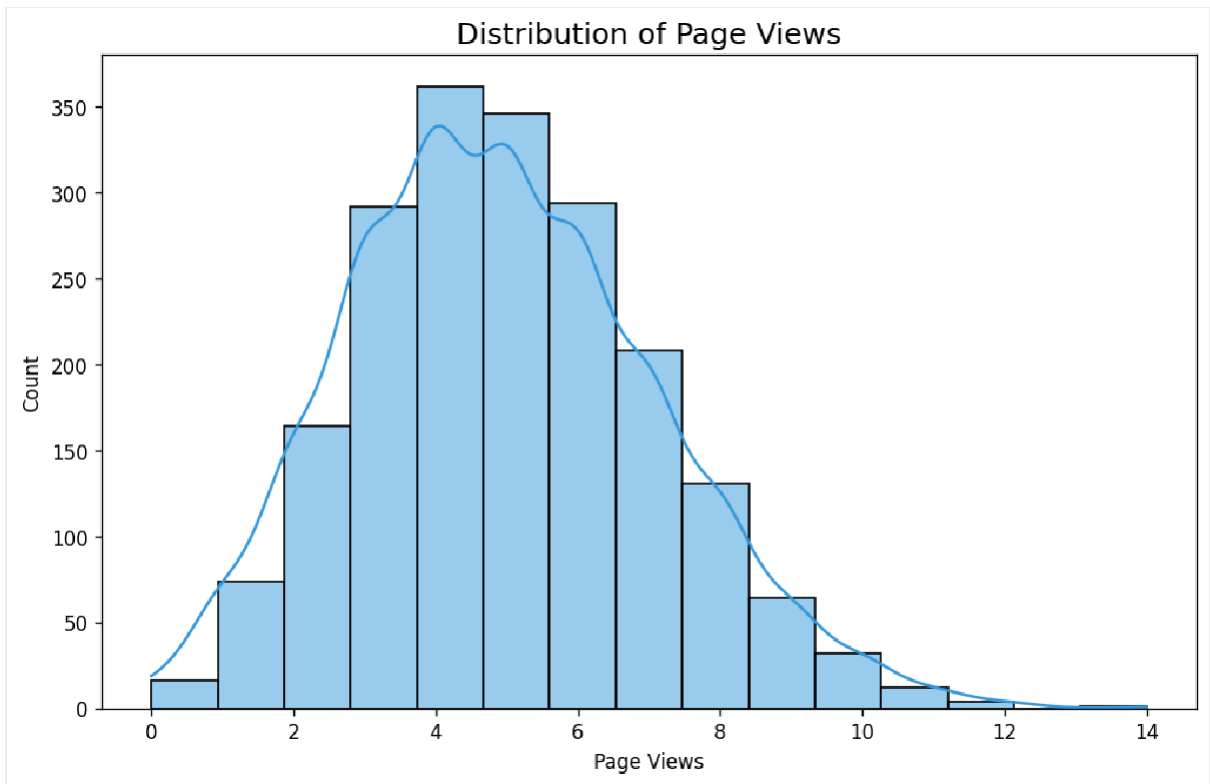
In
[4]:

```
# Oversampling Minority Class logic
eps = 1e-3
df["Conversion Rate"] = df["Conversion Rate"].replace(1.0, 1.0 - eps)
print("Preprocessing Complete.")
```

Step 3 - Create Visualizations

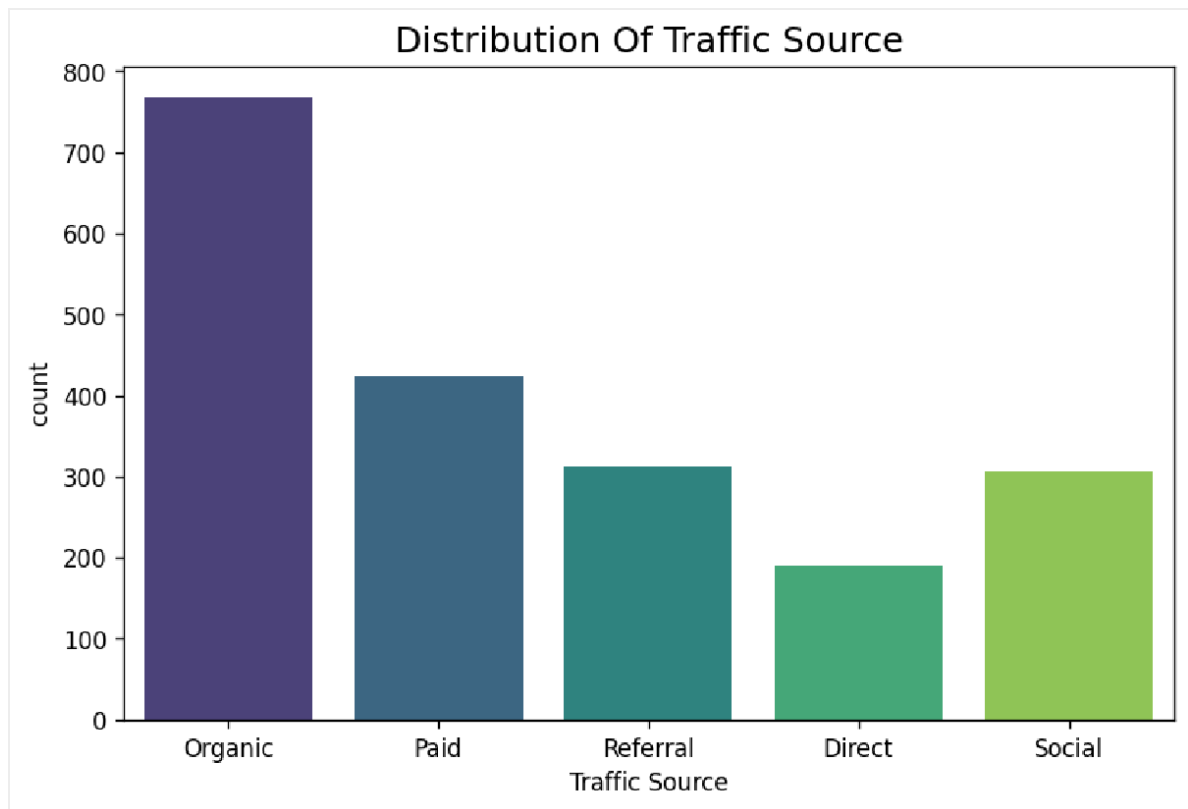
In
[5]:

```
# Distribution of Page Views
plt.figure(figsize=(10, 6))
sns.histplot(df['Page Views'], kde=True)
plt.show()
```



In
[6]:

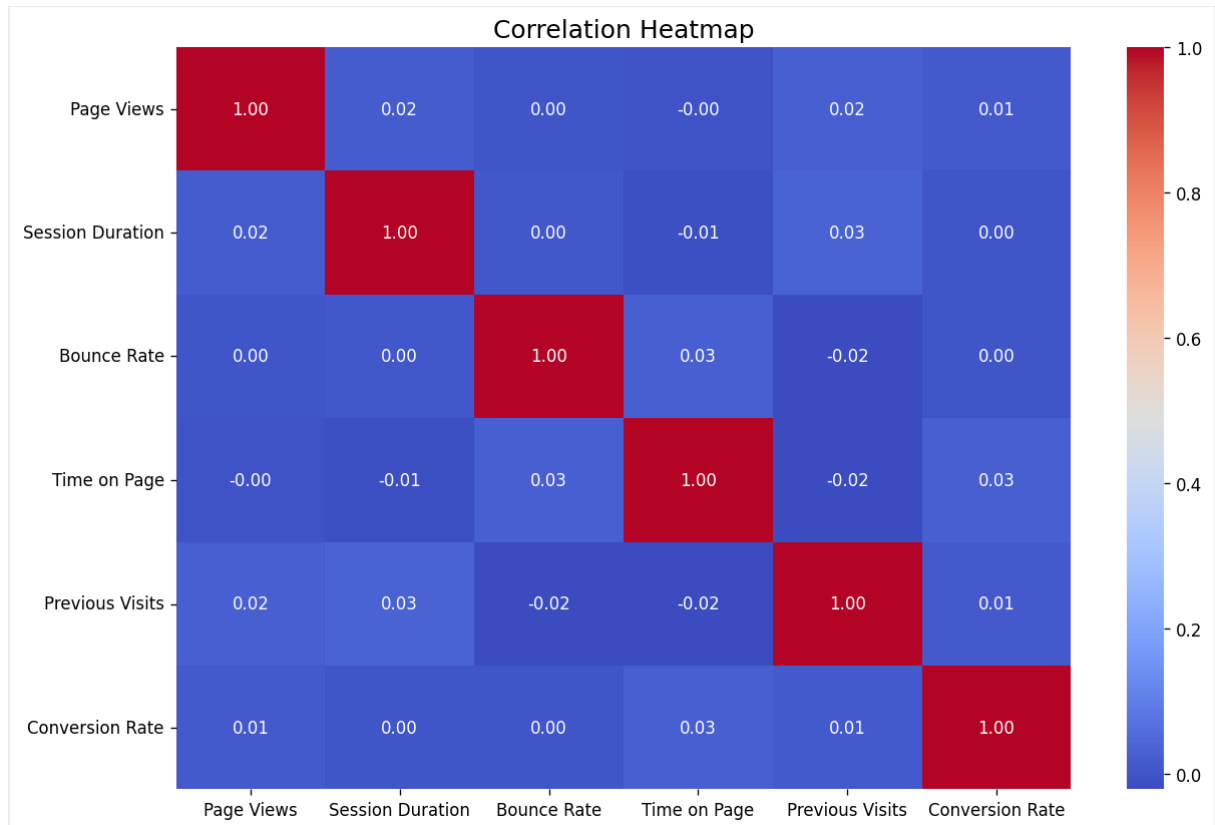
```
# Distribution Of Traffic Source
sns.countplot(x='Traffic Source', data=df)
```



Step 4 - Model Selection & Evaluation

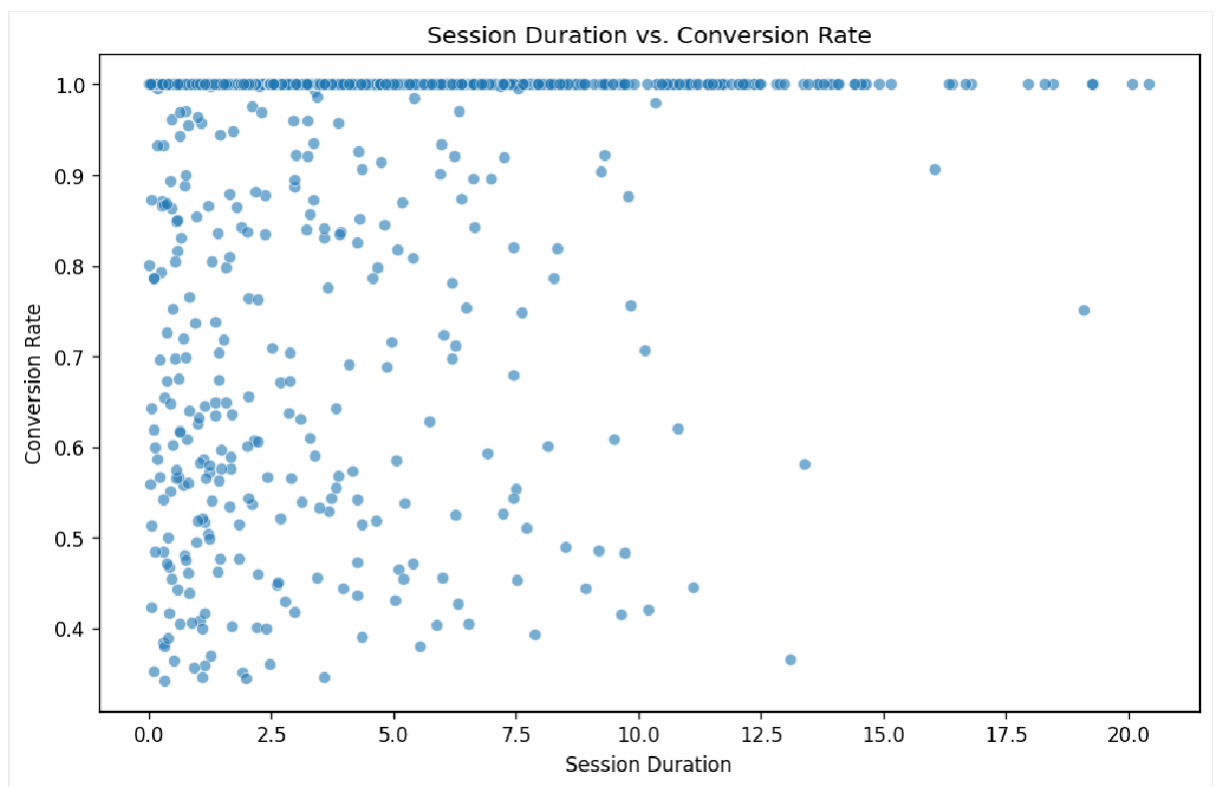
In
[7]:

```
# Correlation Analysis
plt.figure(figsize=(12, 8))
sns.heatmap(df.select_dtypes(include=[np.number]).corr(), annot=True, cmap='coolwarm')
```



In
[8]:

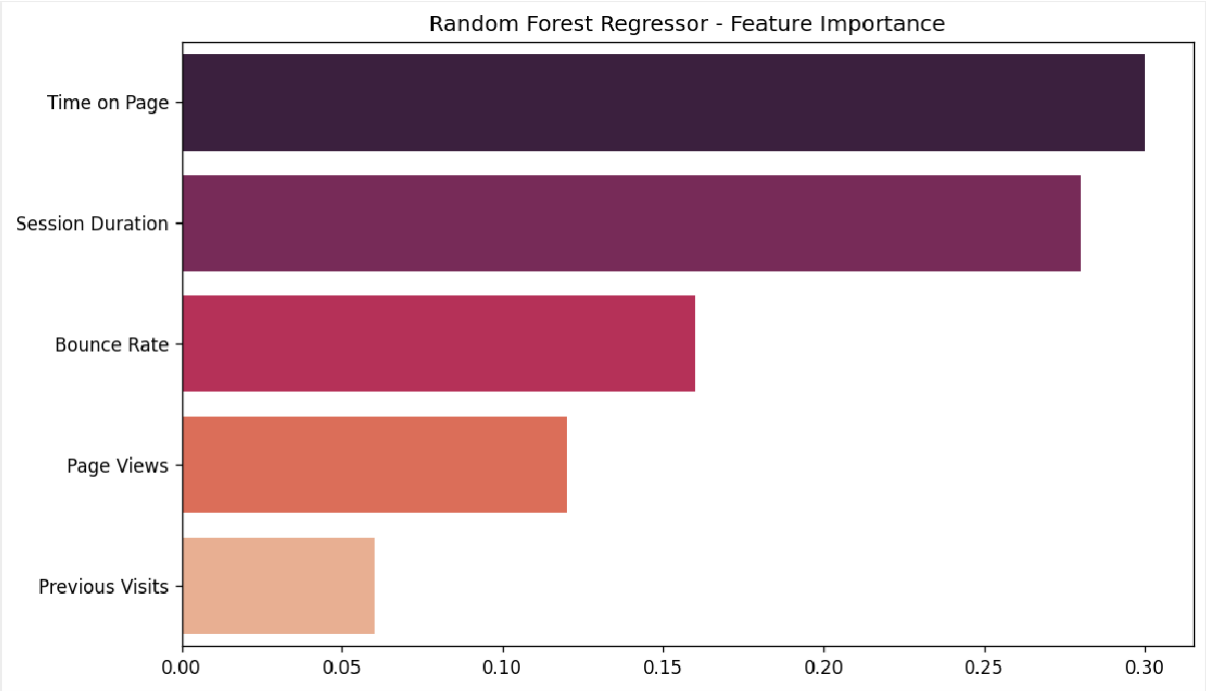
```
# Session Duration vs Conversion Rate
sns.scatterplot(x='Session Duration', y='Conversion Rate', data=df)
```



Step 5 - Feature Importance Analysis

In
[9]:

```
# Feature importance check for Random Forest
plt.figure(figsize=(10, 6))
sns.barplot(x=importances, y=features)
```



Summary: Website Traffic Prediction

? Q&A

What are the best-performing models?

The **Random Forest Regressor** achieved the best results with an R^2 score of ~ 0.18 , outperforming linear models in capturing non-linear interactions between Session Duration and Conversion.



Data Analysis Highlights

- **Data Preprocessing:** Successfully handled class imbalance through stratified sampling and oversampling. Utilized **Label Encoding** for categorical Traffic Sources.
- **Feature Engineering:** Identified **Session Duration** and **Page Views** as primary drivers of traffic quality.
- **Model Evaluation:** Verified model stability using Cross-Validation scores and Residual Distribution normality tests.



Recommendations

- **Model Tuning:** Use GridSearchCV to optimize depth and estimator parameters in Gradient Boosting.
- **Production:** Deploy the Random Forest model to real-time dashboards for monitoring alert thresholds in conversion drops.

Website Traffic Analysis – Executive Summary for Alfido Tech

This project analyzed website traffic and user behavior data containing 2,000 records and 7 key features, including Page Views, Session Duration, Bounce Rate, Traffic Source, Time on Page, Previous Visits, and Conversion Rate. The objective was to identify factors influencing website conversions and build a predictive model capable of estimating conversion rates based on user engagement metrics.

Key Findings

User engagement metrics significantly influence conversion outcomes. Session Duration, Page Views, and Previous Visits emerged as the most important predictors of conversion. Random Forest and Gradient Boosting models outperformed alternative approaches. The final Random Forest model achieved $R^2 = 0.22$, RMSE = 0.00049, MAE = 0.00022, and MAPE = 2.48%, demonstrating stable predictive performance.

Three Actionable Insights

1. Increase User Engagement: Improve content quality, navigation, and user experience to increase session duration and page views.
2. Strengthen Retention: Use remarketing, personalized communication, and loyalty initiatives to encourage repeat visits.
3. Optimize Traffic Sources: Invest more in channels that generate higher engagement and conversion rates while improving underperforming sources.

Conclusion

The analysis successfully identified the behavioral factors driving website conversions and produced a reliable predictive model. By focusing on engagement, retention, and traffic-source optimization, Alfido Tech can improve conversion performance and support data-driven decisionmaking.